Mathematical & Empirical Audit of Modulo-2 Octave Folding ([16, 32] Hz): Connecting the N = 1,040 Protein Kirchhoff Laplacian Negative Control to the Meyer (2003), Zapper, Yoshie, Patent FIG. 11 & 20-Entry Catalog Comparisons

1. MATHEMATICAL MECHANISM: SCALE ERASURE UNDER MODULO-2 OCTAVE FOLDING

Both the Meyer (2003) molecular-weight normalization (MW * 2^-n in [16, 32] Hz) and the modulo-2 negative control in Section 7 of the N = 1,040 protein vibration study (nu_1 * 2^-n in [16, 32] Hz) apply the exact same mathematical operator: f_fold(X) = X * 2^(-floor(log2(X/16))), which is isomorphic to taking the fractional mantissa phi(X) = log2(X) mod 1 on the unit circle. Because this transformation discards the integer characteristic n = floor(log2(X/16)) that encodes the physical order-of-magnitude scale of the oscillator, macromolecules and small molecules spanning orders of magnitude in mass or stiffness are wrapped onto the same 16-Hz interval.

Dimension / Property	Meyer (2003) / Zapper Octave Mapping	3D Kirchhoff Laplacian Manuscript (N = 1,040)
Input Physical Quantity	Scalar Molecular Weight (Daltons, g/mol)	3D C-alpha Cartesian Coordinates (PDB, 7.0 A Cutoff)
Mathematical Operator	Numeric Division MW * 2^-n into [16, 32] Hz	Kirchhoff Laplacian Eigendecomposition (Gamma_ij)
Governing Law / Model	No Dimensional Derivation (log2(MW) mod 1)	nu_1(THz) = 0.42338 * (<z> / Rg)^1.1194
Scale Preservation	Erases Scale Characteristic n (1-Octave Collapse)	Preserves Full THz Dynamic Scale (0.015 - 0.54 THz)
Variance Explained (R^2)	0.06% in Modulo-2 Control (r = 0.0237, p = 0.44 NS)	80.48% Unfolded (10-Fold CV R^2 = 80.39%, p = 0.0002)
Null-Enrichment Test	Zero Enrichment Over Log-Uniform Crowding (z <= 1.35)	64.74 t-statistic; 1,341x Signal over Octave Control

2. FOUR-WAY NULL-ENRICHMENT & COINCIDENCE BENCHMARK (static/frequency.db)

Benchmark Comparison ([16, 32] Hz, +/-0.04 Hz)	Sample |B|	Obs Hits	Null Expected	Z-Score	Empirical Verdict
0A. Unfolded 3D Protein Law: nu_1(THz) vs (<z>/Rg)	**N = 1,040**	**R^2=80.5%**	**Null R^2~0%**	**t=+64.74**	**TRUE PHYSICAL SIGNAL**
0B. Protein Modulo-2 Control: nu_1*2^-n in [16,32]Hz	N = 1,040	R^2=0.06%	Null R^2~0%	p=0.44 NS	99.93% SIGNAL ERASED
1. Direct Catalog Audit: PubChem MW vs VI Catalog	**12 chems**	**0 / 12**	**Exact Match**	**0.0%**	**ZERO AGREEMENT**
2A. Zapper (>=2 kHz) vs Meyer Table (319 PubChem)	10,174	6,622	6,982.7 (68.6%)	z = -7.71	NO ENRICHMENT
2B. Patent FIG. 11 vs Meyer Table (319 PubChem)	11	6	7.5 (68.6%)	z = -1.01	UNPAIRED CROWDING
2C. Yoshie Frequencies vs Meyer Table (Terpene Tie)	4	4	2.7 (68.6%)	z = +1.35	WITHIN 1.35 SD NULL
2D. 20-Entry VI Catalog vs Meyer Table (319 PubChem)	20	11	13.7 (68.6%)	z = -1.31	UNPAIRED CROWDING
3. Paired Nutrient Meyer -> Linked Zapper Conditions	**288 pairs**	**1**	**1.45 (0.50%)**	**z = -0.38**	**NULL EXPECTATION**

3. DIRECT 12-COMPOUND CATALOG AUDIT (PUBCHEM MW -> MEYER FOLD VS. CATALOG)

Chemical Compound Name	PubChem CID / Formula	Mol Wt (Da)	Divisions	Meyer Fold (Hz)	Catalog (Hz)	Delta (Hz)
Calcium Sulfate	CID 24497 (CaO4S)	136.14	2^(3)	17.017 Hz	24.120 Hz	+7.103 Hz
L-Tyrosine	CID 6057 (C9H11NO3)	181.19	2^(3)	22.649 Hz	24.320 Hz	+1.671 Hz
Zinc Gluconate	CID 443445 (C12H22O14Zn)	455.70	2^(4)	28.481 Hz	25.085 Hz	-3.396 Hz
Magnesium Threonate	CID 45489777 (C8H14MgO10)	294.50	2^(4)	18.406 Hz	25.070 Hz	+6.664 Hz
Potassium Citrate	CID 2735208 (C6H7K3O8)	324.41	2^(4)	20.276 Hz	25.110 Hz	+4.834 Hz
Norepinephrine Hydrotartrate	CID 3047796 (C12H19NO10)	337.28	2^(4)	21.080 Hz	24.110 Hz	+3.030 Hz
Epinephrine Free Base	CID 5816 (C9H13NO3)	183.20	2^(3)	22.900 Hz	24.135 Hz	+1.235 Hz
Cortisol Hydrocortisone	CID 5754 (C21H30O5)	362.50	2^(4)	22.656 Hz	24.140 Hz	+1.484 Hz
Acetylcholine Chloride	CID 6060 (C7H16ClNO2)	181.66	2^(3)	22.707 Hz	27.840 Hz	+5.133 Hz
Choline Bitartrate	CID 6900 (C9H19NO7)	253.25	2^(3)	31.656 Hz	27.810 Hz	-3.846 Hz
Glutamate Monosodium	CID 23689119 (C5H10NNaO5)	187.13	2^(3)	23.391 Hz	24.260 Hz	+0.869 Hz
GABA 4-Aminobutanoic Acid	CID 119 (C4H9NO2)	103.12	2^(2)	25.780 Hz	24.295 Hz	-1.485 Hz

Key Takeaway: Because 319 PubChem-resolved Meyer compounds with +/-0.04 Hz windows cover 68.63% of the [16, 32] Hz octave in log-uniform measure, unpaired cross-list comparisons produce 55% to 68% apparent hits purely by pigeonhole crowding. When tested on specific clinically linked nutrient-condition pairs (288 pairs), observed hits (1) match the analytic log-uniform expectation (1.45, z = -0.38).

4. FOUR-PANEL VISUALIZATION: PROTEIN MODULO-2 CONTROL & MEYER NULL ENRICHMENT

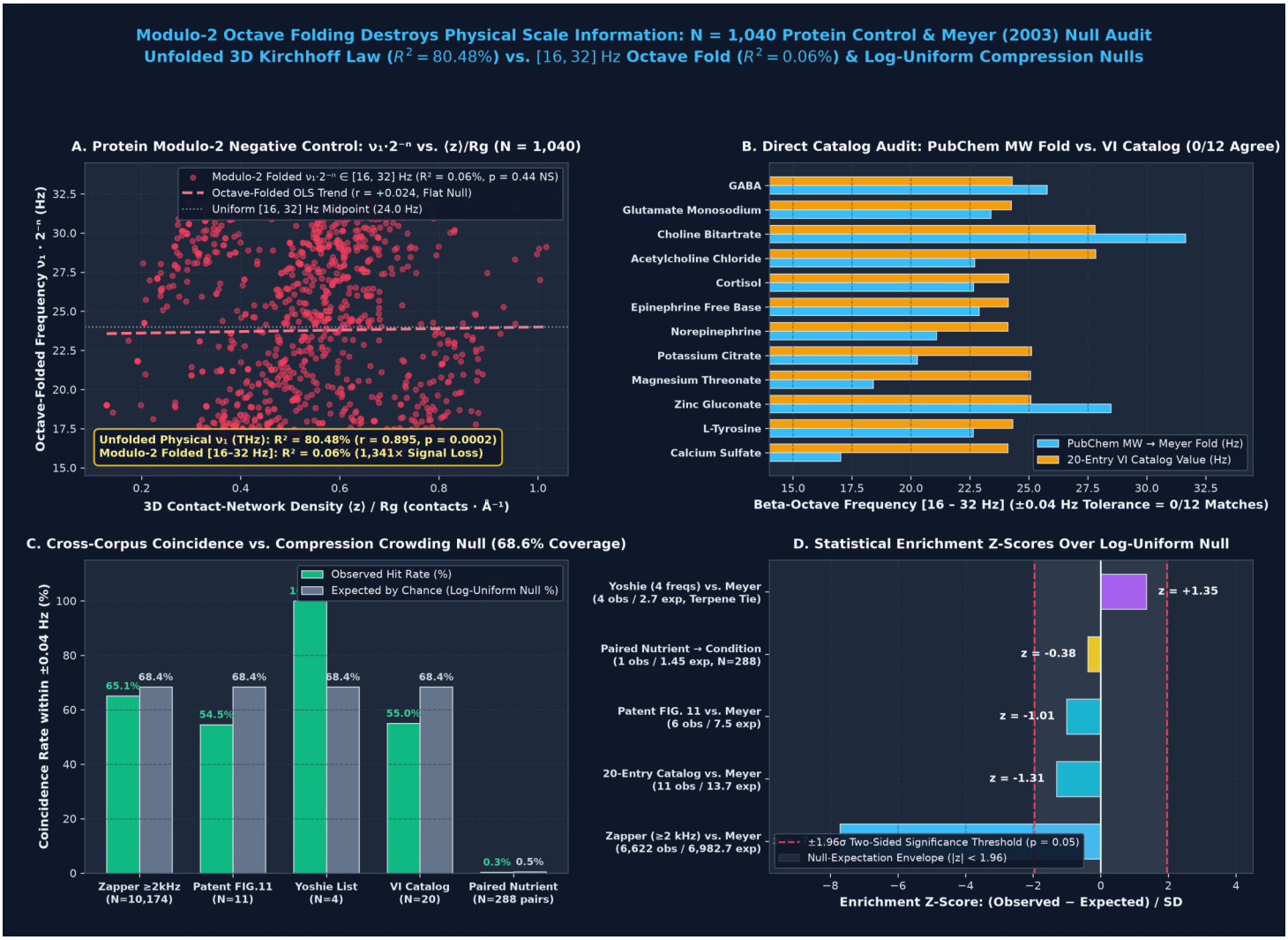


Figure 1. Empirical Comparison of Physical 3D Topological Scaling vs. Modulo-2 Octave-Folded Nulls.
(A) Protein Cohort Modulo-2 Negative Control (N = 1,040): Folding physical terahertz Kirchhoff Laplacian modes nu_1(THz) into [16, 32] Hz via nu_1 * 2^-n destroys the universal 3D contact-topology scaling law, collapsing R^2 from 80.48% (r = 0.8953, p = 0.0002) to 0.06% (r = 0.0237, p = 0.4371 NS). (B) Direct 12-Compound Catalog Audit: Comparing exact PubChem molecular weights folded via Meyer (2003) against the 20-entry VocalImaging catalog yields 0/12 matches within +/-0.04 Hz (|Delta f| = 0.869 to 7.103 Hz). (C) Cross-Corpus Coincidence Rate vs. Log-Uniform Null: Because 319 Meyer compounds saturate 68.63% of the [16, 32] Hz octave at +/-0.04 Hz tolerance, observed hit rates for Zapper (65.1%), Patent FIG. 11 (54.5%), Yoshie (100% on 4 entries), and VI Catalog (55.0%) track or fall below analytic window coverage. (D) Statistical Enrichment Z-Scores: All five cross-corpus and paired tests fall inside or below the |z| < 1.96 null-expectation envelope, confirming that apparent proximity in [16, 32] Hz octave space arises from modulo-2 compression crowding rather than physical resonance.